From Matchings to Max-Flow Min-Cut

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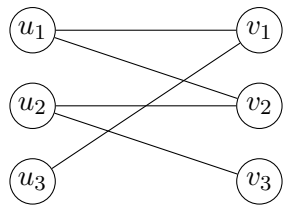
1 Matchings and Alternating Paths

1.1 Bipartite Graphs

A **bipartite graph** $G = (U \cup V, E)$ has two disjoint vertex sets with edges only between them.



The fundamental question is: *how many independent pairs can we match?*

1.2 Matchings

A **matching** $M \subseteq E$ is a set of edges with no shared endpoints. A vertex is **matched** if it belongs to some edge in M; otherwise it is **free**. A **maximum matching** maximises $|M|$. A **perfect matching** matches every vertex ($|M| = |U| = |V|$).

1.3 Alternating and Augmenting Paths

Given a matching M, an **alternating path** alternates between edges in M and edges not in M. An **augmenting path** is an alternating path whose two endpoints are both free.

If an augmenting path exists, flipping every edge along it (matched $\leftrightarrow$ unmatched) yields a matching of size $|M| + 1$.

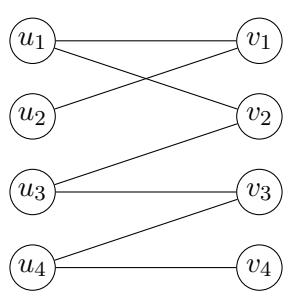
Theorem 1 (Berge, 1957)**.** *A matching M is maximum if and only if there is no augmenting path with respect to M.*

1.4 Algorithm

Berge's theorem gives an algorithm: start with $M = \emptyset$, repeatedly search for an augmenting path P (via BFS or DFS from free vertices), and set $M \leftarrow M \oplus P$ (symmetric difference). Stop when no augmenting path exists. Each augmentation increases $|M|$ by one, so the algorithm terminates after at most $\lfloor |V|/2 \rfloor$ iterations.

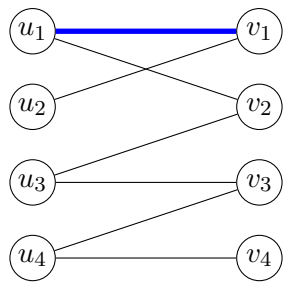
1.5 Example: Augmenting Paths Step by Step

Consider the bipartite graph with $U = \{u_1, u_2, u_3, u_4\}$, $V = \{v_1, v_2, v_3, v_4\}$ and edges:



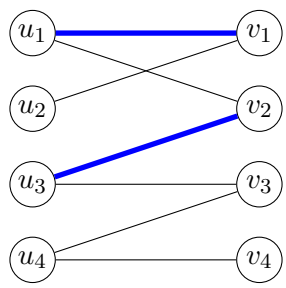
Note that u_2 has only one neighbour (v_1), so any perfect matching must assign $u_2 \to v_1$. The algorithm does not know this in advance.

Step 1. $M = \emptyset$. Search from u_1 (free): find v_1 (free). Augmenting path u_1–v_1. Augment.



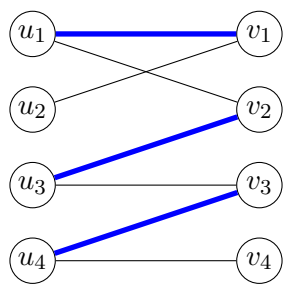
$M = \{(u_1, v_1)\}$, $|M| = 1$.

Step 2. Search from u_3 (free): find v_2 (free). Augmenting path u_3–v_2. Augment.



$M = \{(u_1, v_1), (u_3, v_2)\}$, $|M| = 2$.

Step 3. Search from u_4 (free): find v_3 (free). Augmenting path u_4–v_3. Augment.

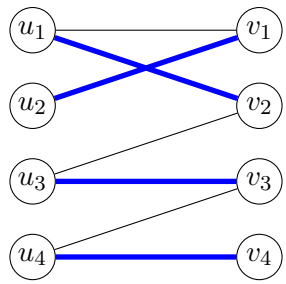


$M = \{(u_1, v_1), (u_3, v_2), (u_4, v_3)\}$, $|M| = 3$.

Step 4. The only free vertex in U is u_2. Its sole neighbour is v_1, which is matched to u_1. No free vertex is directly reachable, so BFS must follow alternating edges through the matching:

$$\underset{\text{free}}{u_2} \xrightarrow{\text{unmatched}} v_1 \xrightarrow{\text{matched}} u_1 \xrightarrow{\text{unmatched}} v_2 \xrightarrow{\text{matched}} u_3 \xrightarrow{\text{unmatched}} v_3 \xrightarrow{\text{matched}} u_4 \xrightarrow{\text{unmatched}} \underset{\text{free}}{v_4}$$

This is an augmenting path of length 7 (four unmatched edges, three matched edges). Flipping every edge along the path yields:



$M = \{(u_2, v_1), (u_1, v_2), (u_3, v_3), (u_4, v_4)\}$, $|M| = 4$. This is a perfect matching.

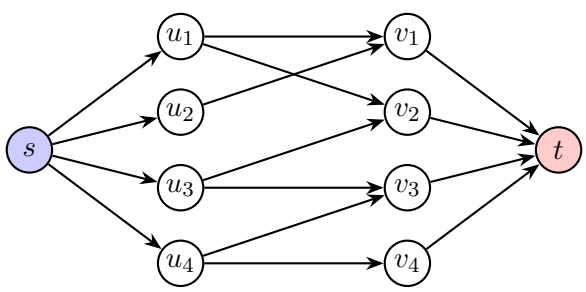
Step 5. No free vertices remain. No augmenting path exists. By Berge's theorem, M is a maximum matching.

The key observation is Step 4: the algorithm rerouted three existing matched edges through a single augmenting path. The "cascade" $v_1 \to u_1 \to v_2 \to u_3 \to v_3 \to u_4$ displaced every previous assignment to make room for u_2. Without this ability to undo prior choices, a greedy algorithm would be stuck at $|M| = 3$.

2 From Matchings to Network Flow

2.1 Adding a Source and Sink

A bipartite matching problem is a network flow problem in disguise. We convert the same graph from Section 1 into a flow network by adding a source s with capacity-1 edges to every $u \in U$, a sink t with capacity-1 edges from every $v \in V$, and directing the original edges from U to V with capacity 1. All capacities are 1.



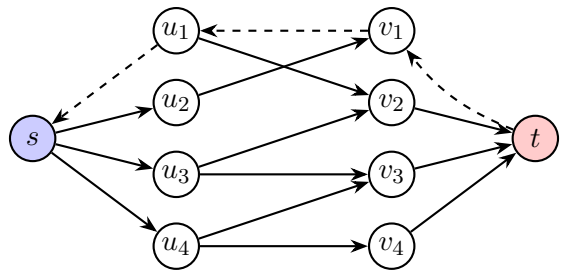
An integer maximum flow in this network corresponds exactly to a maximum matching: each unit of flow along $s \to u_i \to v_j \to t$ represents the matched edge (u_i, v_j).

2.2 Ford–Fulkerson on the Same Graph

We now trace the Ford–Fulkerson algorithm on this network. At each step we show the residual graph, find an augmenting $s \to t$ path, and push one unit of flow. Edges labelled with their residual capacity; forward edges are solid, backward edges are dashed.

Iteration 1. All flows are zero. The residual graph equals the original network. BFS from s finds the path $s \to u_1 \to v_1 \to t$ with bottleneck $\delta = 1$. Push 1 unit.

Flow: $f(s, u_1) = f(u_1, v_1) = f(v_1, t) = 1$. Total $|f| = 1$.



Note the backward edges $u_1 \dashleftarrow v_1$, $s \dashleftarrow u_1$, and $v_1 \dashleftarrow t$: these allow future iterations to undo this flow.

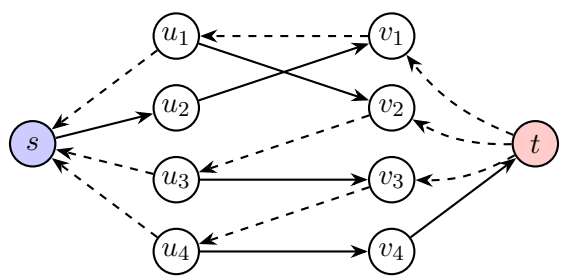
Iteration 2. BFS from s finds $s \to u_3 \to v_2 \to t$. Push 1 unit.

Flow: additionally $f(s, u_3) = f(u_3, v_2) = f(v_2, t) = 1$. Total $|f| = 2$.

Iteration 3. BFS from s finds $s \to u_4 \to v_3 \to t$. Push 1 unit.

Flow: additionally $f(s, u_4) = f(u_4, v_3) = f(v_3, t) = 1$. Total $|f| = 3$.

The residual graph after three iterations:



Iteration 4. BFS from s. The only forward edge from s is $s \to u_2$. From u_2, the only forward edge is $u_2 \to v_1$. But $v_1 \to t$ is saturated (only the backward edge $t \to v_1$ exists). So BFS must use the backward edge $v_1 \to u_1$ (undoing the flow $u_1 \to v_1$), then the forward edge $u_1 \to v_2$, then backward $v_2 \to u_3$, then forward $u_3 \to v_3$, then backward $v_3 \to u_4$, then forward $u_4 \to v_4$, then forward $v_4 \to t$.

The augmenting path in the residual graph:

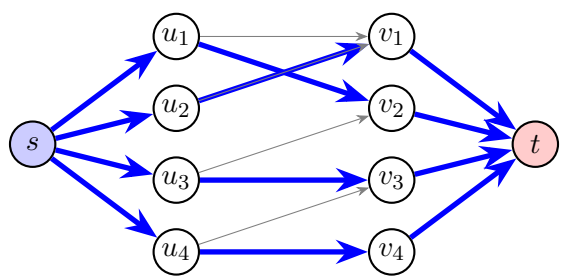
$$s \to u_2 \to v_1 \dashrightarrow u_1 \to v_2 \dashrightarrow u_3 \to v_3 \dashrightarrow u_4 \to v_4 \to t$$

where $\to$ denotes a forward edge and $\dashrightarrow$ a backward edge (undoing existing flow).

Bottleneck $\delta = 1$ (all residual capacities are 1). Push 1 unit along this path. For each forward edge, increase flow by 1; for each backward edge, *decrease* flow by 1 on the reverse arc:

Edge in path	Type	Update
$s \to u_2$	forward	$f(s, u_2) : 0 \to 1$
$u_2 \to v_1$	forward	$f(u_2, v_1) : 0 \to 1$
$v_1 \to u_1$	backward	$f(u_1, v_1) : 1 \to 0$
$u_1 \to v_2$	forward	$f(u_1, v_2) : 0 \to 1$
$v_2 \to u_3$	backward	$f(u_3, v_2) : 1 \to 0$
$u_3 \to v_3$	forward	$f(u_3, v_3) : 0 \to 1$
$v_3 \to u_4$	backward	$f(u_4, v_3) : 1 \to 0$
$u_4 \to v_4$	forward	$f(u_4, v_4) : 0 \to 1$
$v_4 \to t$	forward	$f(v_4, t) : 0 \to 1$

Total $|f| = 4$. The resulting flow (and corresponding matching):



Iteration 5. In the residual graph, all edges from s are saturated (backward only). No $s \to t$ path exists. The algorithm terminates.

The set of vertices reachable from s in the final residual graph is $S^* = \{s\}$ (no forward edge leaves s). The minimum cut is $(\{s\}, V \setminus \{s\})$ with capacity 4, confirming $\max |f| = \min c(S, T) = 4$.

Comparison with Section 1. The flow in iteration 4 corresponds exactly to the augmenting path in Step 4 of the matching algorithm. The backward edges in the residual graph play the same role as the matched edges in the alternating path: they let the algorithm *undo* previous flow assignments. Ford–Fulkerson on the flow network is the augmenting-paths algorithm in disguise.

2.3 Applications

Network flow models appear throughout combinatorics and applied mathematics: bipartite matching (job assignment, scheduling), transportation and logistics, telecommunications routing, image segmentation (min-cut separates foreground from background), baseball elimination, and—for our purposes—assigning components to lines.

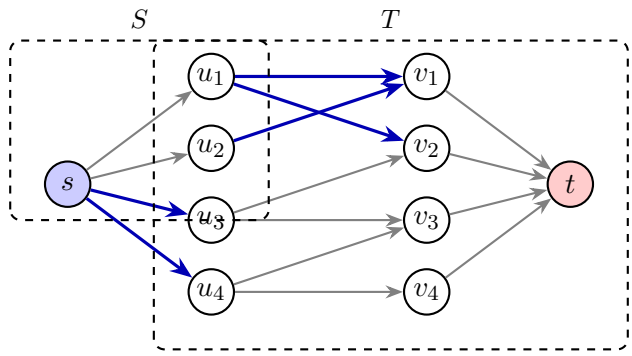
3 Beyond Bipartite Graphs

The bipartite flow network above has a simple layered structure, but flow and cuts apply to arbitrary directed graphs. The reason is that every cut in a general graph *already is* a bipartite bottleneck.

3.1 Every Cut Is a Bipartite Interface

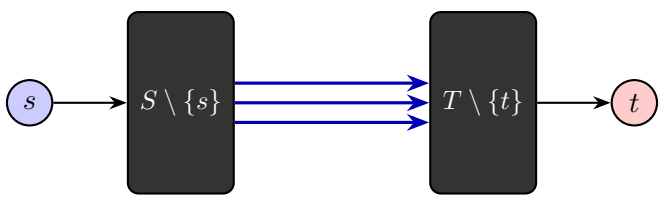
Take any s-t cut (S, T) in a directed graph. The edges crossing from S to T form a bipartite structure: each has one endpoint in S and one in T. Every unit of flow from s to t must cross this interface.

Consider the cut $S = \{s, u_1, u_2\}$, $T = \{u_3, u_4, v_1, v_2, v_3, v_4, t\}$ in our flow network:



The five bold edges are the bipartite crossing (capacity 5); the gray edges are internal plumbing. No matter what happens inside S or T, flow from s to t cannot exceed 5 through this cut.

From the perspective of the cut, the internal structure on each side is irrelevant—only the crossing matters:



bipartite crossing: capacity = 5

The black boxes hide arbitrarily complex subgraphs. All that determines the throughput bound is the capacity of the edges between them.

3.2 Internal Edges Are Lossless Plumbing

Flow conservation forces every internal node to output exactly what it receives. The subgraph inside S merely *routes* flow from s to the cut boundary—it cannot create or destroy flow. Symmetrically, the subgraph inside T routes flow from the boundary to t.

In our example, u_1 and u_2 sit inside S. They receive flow from s and pass all of it across the cut to v_1 and v_2. They are plumbing, not bottlenecks.

Therefore, for any cut (S, T):

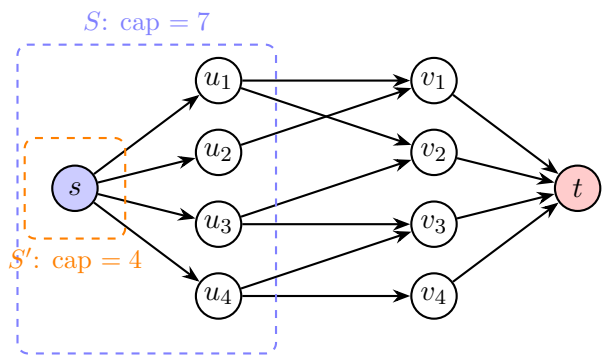
$$|f| = \underbrace{\sum_{u \in S,\, v \in T} f(u, v)}_{\text{flow across the cut}} - \underbrace{\sum_{u \in T,\, v \in S} f(u, v)}_{\text{flow back across}} \leq c(S, T).$$

The bottleneck is always at the bipartite crossing, never inside either side.

3.3 Nested Cuts Capture Internal Limits

An immediate objection: what if the edges *inside* S are too narrow to deliver flow to the cut boundary? Then a tighter cut exists that captures the true bottleneck.

Compare two cuts in our flow network:



The wider cut $S = \{s, u_1, u_2, u_3, u_4\}$ has 7 edges crossing to V (capacity 7). But s has only 4 outgoing edges, so the internal plumbing cannot deliver more than 4 to the boundary. The tighter cut $S' = \{s\}$ exposes the real bottleneck: capacity 4.

Internal limitations never hide from cuts—they just mean the minimum cut is closer to s (or t). Every general graph is a chain of bipartite bottlenecks; the minimum cut finds the tightest one.

4 Formal Definitions: Flows and Cuts

We now make the preceding intuition precise.

4.1 Flow Networks

Definition 1 (Network)**.** *A **network** is a directed graph $G = (V, E)$ with a non-negative capacity function $c : E \to \mathbb{R}_{\geq 0}$ and no multiple arcs. Without loss of generality, if $(u, v) \in E$ then $(v, u) \in E$ as well (add it with $c(v, u) = 0$ if absent).*

Definition 2 (Flow network)**.** *If two vertices are distinguished—a **source** s and a **sink** t—then (G, c, s, t) is a **flow network**.*

4.2 Flows

Definition 3 (Flow)**.** *A **flow** is a function $f : E \to \mathbb{R}_{\geq 0}$ satisfying:*

1. ***Capacity**: $0 \leq f(u, v) \leq c(u, v)$ for all $(u, v) \in E$.*
2. ***Conservation**: for every $v \in V \setminus \{s, t\}$,*

$$\sum_{u:(u,v)\in E} f(u, v) = \sum_{w:(v,w)\in E} f(v, w).$$

The **value** of a flow is $|f| = \sum_{v:(s,v)\in E} f(s, v) - \sum_{u:(u,s)\in E} f(u, s)$.

4.3 Cuts

Definition 4 (s-t cut)**.** *An s-t* ***cut*** *(S, T) is a partition $V = S \cup T$ with $s \in S$ and $t \in T$. Its* ***capacity*** *is*

$$c(S, T) = \sum_{\substack{u \in S,\, v \in T \\ (u,v) \in E}} c(u, v).$$

A cut is a bottleneck: it bounds the maximum flow that can cross from the s-side to the t-side.

5 The Max-Flow Min-Cut Theorem

5.1 Statement

Theorem 2 (Ford–Fulkerson, 1956)**.** *In any flow network (G, c, s, t),*

$$\max_f |f| = \min_{(S,T)} c(S, T).$$

The following are equivalent: (i) f is a maximum flow; (ii) there is no augmenting path in the residual graph G_f; (iii) $|f| = c(S, T)$ for some cut (S, T).

5.2 Residual Graph

Given flow f, the **residual graph** $G_f = (V, E_f)$ contains a **forward edge** (u, v) with capacity $c_f(u, v) = c(u, v) - f(u, v)$ whenever this is positive, and a **backward edge** (v, u) with capacity $c_f(v, u) = f(u, v)$ whenever $f(u, v) > 0$. An **augmenting path** is any $s \to t$ path in G_f. Backward edges allow the algorithm to undo previously pushed flow—the same mechanism as augmenting paths in matching.

5.3 Proof of Weak Duality

Claim. For any flow f and any cut (S, T), $|f| \le c(S, T)$.

Proof. By flow conservation the net flow leaving S equals $|f|$:

$$|f| = \sum_{u \in S,\, v \in T} f(u, v) - \sum_{u \in T,\, v \in S} f(u, v) \le \sum_{u \in S,\, v \in T} f(u, v) \le \sum_{u \in S,\, v \in T} c(u, v) = c(S, T).$$

The first inequality drops a non-negative term; the second uses the capacity constraint. □

5.4 Proof of Strong Duality

Claim. There exist f^* and (S^*, T^*) with $|f^*| = c(S^*, T^*)$.

Proof. Run Ford–Fulkerson until no augmenting path exists; call the resulting flow f^*. Define

$$S^* = \{v \in V : v \text{ reachable from } s \text{ in } G_{f^*}\}, \qquad T^* = V \setminus S^*.$$

Since $t \notin S^*$, this is a valid cut. For every edge (u, v) with $u \in S^*$, $v \in T^*$ we have $f^*(u, v) = c(u, v)$—otherwise v would be reachable via a forward edge. For every edge (v, u) with $v \in T^*$, $u \in S^*$ we have $f^*(v, u) = 0$—otherwise v would be reachable via a backward edge. Therefore

$$|f^*| = \sum_{u \in S^*,\, v \in T^*} f^*(u, v) - \sum_{v \in T^*,\, u \in S^*} f^*(v, u) = c(S^*, T^*) - 0 = c(S^*, T^*).$$

By weak duality f^* is maximum and (S^*, T^*) is minimum. □

6 Algorithm and Examples

6.1 Ford–Fulkerson

Input: Flow network (G, c, s, t).
Output: Maximum flow f.

1. Set $f(u, v) \leftarrow 0$ for all $(u, v) \in E$.
2. While there exists an augmenting path P in G_f:
   (a) Let $\delta = \min_{(u,v)\in P} c_f(u, v)$.
   (b) For each forward edge $(u, v) \in P$, set $f(u, v) \leftarrow f(u, v) + \delta$.
   (c) For each backward edge $(v, u) \in P$, set $f(v, u) \leftarrow f(v, u) - \delta$.
3. Return f.

The **Edmonds–Karp** variant uses BFS to find the shortest augmenting path, giving worst-case complexity $O(VE^2)$.

6.2 Example

We trace the algorithm on a small network: start with zero flow, find an augmenting s-t path in the residual graph, push flow along it, update the residual graph (forward edges shrink, backward edges appear), and repeat. When no path exists, the vertices reachable from s in the final residual graph form S^*; the cut $(S^*, V \setminus S^*)$ is the minimum cut.

7 Summary and Related Results

The max-flow min-cut theorem is the cornerstone of a family of duality results in combinatorics.

7.1 Hall's Theorem

Theorem 3 (Hall, 1935)**.** *A bipartite graph $G = (U \cup V, E)$ has a matching saturating every vertex in U iff $|N(S)| \geq |S|$ for all $S \subseteq U$.*

Hall's condition is equivalent to every s-t cut having capacity at least $|U|$ in the corresponding flow network.

7.2 Birkhoff–von Neumann Theorem

Theorem 4 (Birkhoff, 1946; von Neumann, 1953)**.** *Every doubly stochastic matrix is a convex combination of permutation matrices.*

A doubly stochastic matrix has non-negative entries with rows and columns summing to 1. The theorem says the LP relaxation of bipartite matching has no integrality gap: any fractional matching decomposes into integer matchings.

7.3 König–Egerváry Theorem

Theorem 5 (König, 1931; Egerváry, 1931)**.** *In a bipartite graph, the size of a maximum matching equals the size of a minimum vertex cover.*

A vertex cover is a set of vertices touching every edge. The minimum vertex cover corresponds to a minimum cut in the flow network.

7.4 Menger's Theorem

Theorem 6 (Menger, 1927)**.** *The maximum number of vertex-disjoint s-t paths equals the minimum number of vertices (other than s, t) whose removal disconnects s from t.*

The edge-disjoint version—max edge-disjoint paths equals min edge cut—is a direct corollary of max-flow min-cut with unit capacities.

7.5 Unifying View

Result	**Says**
Max-flow min-cut	max throughput = min bottleneck
Hall	perfect matching $\iff$ no small neighbourhood
Birkhoff–von Neumann	fractional matching = integer matching
König–Egerváry	max matching = min vertex cover
Menger	max disjoint paths = min separator

All are facets of the same duality: **what you can push through a network equals what it takes to block it**.